Worksheet 4C: Advanced Rules — Derivatives and Their Integral Inverses

What This Worksheet Is About

Each derivative rule has an integral counterpart: - **Product Rule Integration by Parts** - **Chain Rule Substitution (u-sub)**

These are inverse operations!

Part A: The Product Rule (12 problems)

$$\frac{d}{dx}[u \cdot v] = u' \cdot v + u \cdot v'$$

Find the derivative:

1. $f(x) = x^2(x+1)$

2. $g(t) = t^3(t-4)$

3. $h(p) = (p+2)(p-5)$

4. $y = (2x+1)(3x-2)$

5. $f(m) = m^2(m^2+3m-1)$

6. $g(r) = (r^2-1)(r^2+1)$

7. Verify problem 3 by expanding first.

8. Verify problem 6 by simplifying first: $(r^2-1)(r^2+1) = ?$

With Chain Rule:

9. $h(x) = x^2(x+1)^3$

10. $y = (t+1)^2(t-1)^3$

11. $f(k) = k(2k+1)^4$

12. $g(n) = (n^2+1)^2(n-1)$

Part B: Integration by Parts — Reversing Product Rule (10 problems)

$$\int u\,dv = uv - \int v\,du$$

Connection: If $\frac{d}{dx}[f \cdot g] = f'g + fg'$, then $\int (f'g + fg')dx = fg + C$

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13. The derivative of $x^2(x+1)$ is $3x^2 + 2x$.

Therefore: $\int (3x^2 + 2x)dx =$ _____

14. The derivative of $(x-1)(x+3)$ is $2x + 2$.

Therefore: $\int (2x + 2)dx =$ _____

15. Find $\frac{d}{dx}[(2x+1)(x-2)]$. Then verify $\int$[your answer]dx gives back $(2x+1)(x-2) + C$.

16. Start with $f(x) = x(x+1)^2$. Differentiate, then integrate your answer.

Direct Integration (for comparison):

17. $\int (3x^2 + 2x)dx =$

18. $\int (2x + 2)dx =$

19. Compare: Are problems 13 & 17 the same? Are 14 & 18 the same (up to constants)?

20. Explain why integration is "harder" than differentiation:

- Differentiation: clear rules always work
- Integration: may require _____ or may not have closed form

Preview: Integration by Parts with transcendentals (Worksheet 4D)

21. We'll need by parts for $\int x \cdot e^x dx$ because we can't _____.

22. And for $\int x \cdot \sin(x)dx$ because _____.

Part C: The Quotient Rule (10 problems)

$$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{u'v - uv'}{v^2}$$

Find the derivative:

23. $f(x) = \frac{x}{x+2}$

24. $g(t) = \frac{t^2}{t-1}$

25. $h(p) = \frac{p+1}{p-1}$

26. $y = \frac{x^2+1}{x}$

27. $f(m) = \frac{1}{m^2+1}$

28. $g(r) = \frac{r^3}{r^2+1}$

Quotient Rule vs. Rewriting:

29. Find $\frac{d}{dx}\left[\frac{1}{x^3}\right]$ two ways: - (a) Quotient rule - (b) Rewrite as x^{-3}

30. Find $\frac{d}{dx}\left[\frac{x^4+2x^2}{x^2}\right]$ by simplifying first.

31. Simplify first, then differentiate: $\frac{x^2-1}{x+1}$

32. When MUST you use quotient rule? (When denominator has _____ that don't cancel)

Part D: The Chain Rule (16 problems)

$$\frac{d}{dx}[f(g(x))] = f'(g(x)) \cdot g'(x)$$

"Derivative of outside × derivative of inside"

Find the derivative:

33. $f(x) = (x+3)^4$

34. $g(t) = (2t-1)^5$

35. $h(p) = (p^2+1)^3$

36. $y = (3x^2 - x)^4$

37. $f(m) = (m^3 + m)^2$

38. $g(r) = (1-r^2)^6$

Negative and Fractional Powers:

39. $h(x) = (x+1)^{-1}$

40. $y = (t^2+4)^{-2}$

41. $f(k) = \sqrt{k+5}$

42. $g(n) = \sqrt{n^2+1}$

Verify by Expanding:

43. $(x+2)^3$: Use chain rule, then expand and differentiate. Same answer?

44. $(2x-1)^2$: Both methods.

Combining Rules:

45. $f(x) = x^2(x+1)^3$ (product + chain)

46. $g(t) = \frac{t}{(t+1)^2}$ (quotient + chain)

47. $h(x) = \sqrt{x(x+4)}$ (chain with product inside)

48. $y = \left(\frac{x}{x+1}\right)^3$ (chain on quotient)

Part E: Substitution — Reversing the Chain Rule (16 problems)

$$\int f(g(x)) \cdot g'(x)\,dx = F(g(x)) + C$$

Let $u = g(x)$**, then** $du = g'(x)dx$

Perfect Cases:

49. $\int 3x^2(x^3+1)^4\,dx$

Let $u = x^3+1$, $du = 3x^2dx$

$= \int u^4 du =$

50. $\int 2t(t^2-3)^5\,dt$

51. $\int 4p^3(p^4+2)^{-2}\,dp$

52. $\int \frac{6x^2}{\sqrt{x^3+1}}\,dx$

53. $\int (2m+1)(m^2+m)^3\,dm$

Adjusting Constants:

54. $\int x(x^2+1)^3\,dx$

We need $2xdx$ but have xdx. So: $xdx = \frac{1}{2}(2xdx) = \frac{1}{2}du$

$= \frac{1}{2}\int u^3 du =$

55. $\int t(t^2+5)^4\,dt$

56. $\int \frac{x}{\sqrt{x^2+4}}\,dx$

57. $\int p^2(p^3-1)^2\,dp$

Verify by Differentiating:

58. Check problem 49: Differentiate $\frac{(x^3+1)^5}{5}$ using chain rule.

59. Check problem 54: Differentiate $\frac{(x^2+1)^4}{8}$.

Connection to Chain Rule:

60. Chain rule: $\frac{d}{dx}[(x^2+1)^5] = 10x(x^2+1)^4$

Therefore: $\int 10x(x^2+1)^4 dx =$ _____

61. Chain rule: $\frac{d}{dx}[(t^3-2)^4] = 12t^2(t^3-2)^3$

Therefore: $\int 12t^2(t^3-2)^3 dt =$ _____

62. Explain: Substitution is "chain rule backwards."

More Practice:

63. $\int (3x-2)^4\,dx$

64. $\int (5t+1)^{-3}\,dt$

Part F: Combining Everything (10 problems)

Differentiate:

65. $\frac{x^2}{(x+1)^3}$

66. $x(x^2+1)^{1/2}$

67. $\frac{(x-1)^2}{(x+1)^2}$

Integrate:

68. $\int \frac{2x}{(x^2+1)^3} dx$

69. $\int \frac{x}{\sqrt{x^2+9}} dx$

70. $\int (x+1)(x^2+2x)^4 dx$

Hint: $\frac{d}{dx}(x^2+2x) = 2x+2 = 2(x+1)$

Round-Trips:

71. Start with $(x^2+1)^3$. Differentiate, then integrate back.

72. Start with $x(x+1)^2$. Differentiate, then integrate back.

73. Start with $\frac{x}{x+1}$. Differentiate, then integrate back.

74. Why does "differentiate then integrate" give $f(x)+C$, but "integrate then differentiate" gives exactly $f(x)$?